

Neerav Poluri

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EDUCATION

Purdue University, West Lafayette, IN
Bachelor of Science in Computer Science
GPA: 3.82

Aug. 2025 - May 2028

Illinois Math & Science Academy, Aurora, IL

May 2025

EXPERIENCE

Mechanisms and Robotics Systems Lab - *Undergraduate Researcher* | Purdue University

Sept. 2026 - Present

- Researching the use of pan-tilt-zoom (PTZ) cameras to improve visual perception and data collection for world-action models

ML@Purdue Project team - *Simulation Team* | West Lafayette, IN

Jan. 2026 - Present

- Developing a simulation-first autonomous drone pipeline using ROS 2 and Isaac Sim
- Implementing SLAM, path planning, and reinforcement learning for autonomous navigation

Bechtel Innovation Design Center - *Boilermakery TA* | West Lafayette, IN

Oct. 2025 - Present

- Improved and maintained a Linux-based system for slicing, print data logging, and data backup across a student 3D-printing system supporting 10,000+ print hours per semester
- Repaired and upgraded FDM 3D printers, improving reliability and enabling reservation-free printing for students
- Led workshops on design for manufacturing and FDM optimization

Purdue Boilerbots - *Algorithms Subteam* | West Lafayette, IN

Sept. 2025 - Present

- Improved LiDAR-based localization and OpenCV object-detection systems for an autonomous sentry robot
- Implemented and tuned SLAM, path-planning, and navigation algorithms for autonomous navigation
- Integrated and evaluated ArUco marker detection for robot localization

Setlurs.com & Quasar Design - *Edge Computing & AI Intern* | Chicago, IL

Sept. 2024 - May 2025

- Built computer-vision and machine-learning systems for automated 3D-print monitoring and process control
- Designed an automated print-removal system, integrating computer vision and mechanical systems

PROJECTS

High-Speed 3D Printer Development

Mar. 2022 - Present

- Designed and built high-performance 3D printers from the ground up, including the motion system, frame, hotend, extruder, and parametric external panels
- Experimentally characterized printer behavior and used test data to guide system modifications and optimization
- Applied CAD, DFM, CFD, and rapid prototyping while working with sponsors to source components

Robotic 3-axis Gantry

Apr. 2026

- Designed and built a 3-axis gantry for precise, high-speed motion, integrating mechanical and motion-control systems.
- Engineered an underactuated robotic gripper that uses fewer actuators while improving adaptability to different objects
- Developed OpenCV image-processing pipelines and integrated YOLO for real-time object detection and localization.

Python 3D Printing Slicer

Sept. 2025

- Built a Python-based slicer that converts 3D geometry into layered toolpaths and G-code for additive manufacturing
- Implemented algorithms for perimeter generation, polygon clipping, and infill generation
- Developed toolpath processing to handle holes, disconnected regions, multiple perimeters, and overhangs

SKILLS

- *Languages:* C, C++, Python, Rust, C#, Java
- *Systems:* Linux, Git, NumPy, OpenCV, Machine Learning, Computer Vision
- *Robotics:* ROS 2, Isaac Sim, SLAM, Path Planning, Reinforcement Learning, PID Control
- *CAD/Manufacturing:* SolidWorks, Fusion 360, Onshape, CFD, DFM, FDM 3D Printing, CNC Machining